Program : **Diploma in Instrumentation Engineering**	
Course Code : **5088**	Course Title: **Maintenance & Calibration Workshop III**
Semester : **5**	Credits: **1.5**
Course Category: **Program Core**	
Periods per week : **3 (L:0 T:0 P:3)**	Periods per semester: **45**

Course Objectives:

- To interpret Piping & Instrumentation Diagrams
- To perform servicing of Computers, Flow measuring instruments & Biomedical equipments
- To familiarize Boyle's Apparatus

Course Prerequisites:

Topic	Course code	Course name	Semester
Knowledge of Computer Fundamentals		Introduction to IT system	1
Knowledge of electrical circuits and measurement of electrical parameters		Maintenance and Calibration workshop I	3
Knowledge of Process Control systems		Process Control Instrumentation	4

Course Outcomes:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Implement P&ID for process control application	7	Applying
CO2	Practice servicing of computers and recorders	11	Applying
CO3	Perform calibration and servicing of measuring instruments and final control elements	18	Applying
CO4	Perform servicing of biomedical equipments	6	Applying
	Lab Test	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	2				
CO2	3					2	
CO3	3			2		2	2
CO4	3	2			2		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Implement P&ID for process control application**		
M1.1	Piping & Instrumentation Diagram i) Identify letter, line and instrument symbols in P&ID ii) Read and interpret the information in P&ID and apply it in executing practical work	3	Applying
M1.2	P&ID of Process control loops Prepare P&ID's of process control loops	4	Applying
CO2	**Practice servicing of computers and recorders**		
M2.1	Servicing of Computers i) Dismantle and assemble computer systems. ii) Demonstrate various parts of the system unit and mother board components. iii) Identify various indicators, cables, connectors and ports iv) Test and replace power supply unit v) Test and replace DVD and HDD in the system	5	Applying

M2.2	Servicing of Recorders i) Repair and rectify fault in circular chart recorder. ii) Repair and rectify fault in strip chart recorder.	3	Applying
M2.3	Servicing of Printers i) Identify the parts of ink jet and laser printers. ii) Troubleshoot hardware problems of ink jet and laser printers.	3	Applying
	Lab Test – I	1	
CO3	**Perform calibration and servicing of measuring instruments and final control elements**		
M3.1	Calibrate Flow measuring instruments, I/P and P/I Converters i) Calibrate and test Rotameter. ii) Calibrate and service different flow measuring instruments. iii) Calibrate I/P and P/I converter	3	Applying
M3.2	Weigh Bridge Identify elements in a weighing machine/weigh bridge, Dismantle, Assemble and Calibrate it	3	Applying
M3.3	Flow Control Loop Prepare a flow control loop using flow measuring instruments, indicating and controlling instruments like transmitters, flow restrictors, flow control valve, flow meter totalizes etc with proper fittings and connectors.	3	Applying
M3.4	Control Valve i) Dismantle the control valve with standard tools and equipments. ii) Assemble a pneumatic control valve and test. iii) Install and commission the control valve in a process loop	9	Applying
CO4	**Perform servicing of biomedical equipments**		
M4.1	Servicing of biomedical equipments i) Identify the parts of BP apparatus ii) Rectify the faults in BP apparatus iii) Study and service Diathermy equipments iv) Study and service E.C.G Machine v) Study and service E.E.G Machine	3	Applying

M4.2	Boyle's apparatus and Medical gas pipe line system i) Identify the parts and study the working of Boyle's apparatus ii) Assemble Boyle's apparatus and test the equipment iii) Study and identify the components of medical gas pipe line system.	3	Applying
	Lab Test – II	2	

Text / Reference:

T/R	Book Title/Author
R1	Process control instrumentation technology, Curtis D Johnson
R2	Instrument Engineers Hand book Volume II, Bela G Liptak, Chilton Book Company
R3	Computer based Industrial Control, Krishnakanth
R4	Automatic Process Control, Donald P Eckman
R5	Measurement Systems application and Design 5th Edition, Ernest O Doebelin

Online Resources:

Sl. No	Website Link
1	http://nptel.ac.in/courses/103103037/
2	https://en.wikipedia.org/wiki/Piping_and_instrumentation_diagram
3	https://scienceq.org/wp-content/uploads/2018/10/JAETV7I102.pdf
4	https://processcontrol.tv/tutorials
5	https://www.youtube.com/watch?v=5SmU3rMoHKM